

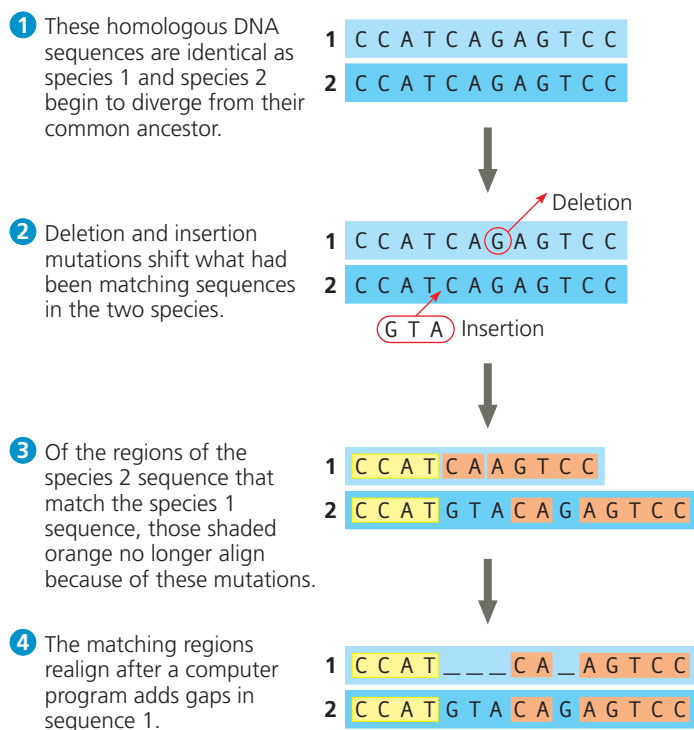
Evaluating Molecular Homologies

Comparing DNA molecules often poses technical challenges for researchers. The first step after sequencing the DNA is to align comparable sequences from the species being studied. If the species are very closely related, the sequences probably differ at only one or a few sites. In contrast, comparable nucleic acid sequences in distantly related species usually have different bases at many sites and may have different lengths. This is because insertions and deletions accumulate over long periods of time.

Suppose, for example, that certain noncoding DNA sequences near a particular gene are very similar in two species, except that the first base of the sequence has been deleted in one of the species. The effect is that the remaining sequence shifts back one notch. A comparison of the two sequences that does not take this deletion into account would overlook what in fact is a very good match. Computer programs can be used to identify such matches by testing possible alignments for comparable DNA segments of differing lengths (**Figure 26.8**).

Such molecular comparisons reveal that many base substitutions and other differences have accumulated in the comparable genes of an Australian “mole” and a golden mole. The many differences indicate that their lineages have diverged greatly since their common ancestor; thus, we say that the living species are not closely related. In contrast, the high degree of gene sequence similarity among the silversword

▼ **Figure 26.8 Aligning segments of DNA.** Systematists search for similar sequences along DNA segments from two species (only one DNA strand is shown for each species). In this example, 11 of the original 12 bases have not changed since the species diverged. Hence, those portions of the sequences still align once the length is adjusted.



▼ **Figure 26.9 A coincidental match.** These two DNA sequences from distantly related species coincidentally share 23% of their bases (those highlighted in yellow).

A C G G A T A G T C C A C T A G G C A C T A
T C A C C G A C A G G T C T T T G A C T A G

WHAT IF? Why might you expect organisms that are not closely related to nevertheless share roughly 25% of their bases?

plants indicates that they are all very closely related, in spite of their considerable morphological differences.

Just as with morphological characters, it is necessary to distinguish homology from analogy in evaluating molecular similarities for evolutionary studies. Two sequences that resemble each other at many points along their length most likely are homologous (see **Figure 26.8**). But in organisms that do not appear to be closely related, the bases that their otherwise very different sequences happen to share may simply be coincidental matches (**Figure 26.9**). Scientists have developed statistical tools that can help distinguish “distant” homologies from such coincidental matches in extremely divergent sequences.

CONCEPT CHECK 26.2

- Decide whether each of the following pairs of structures more likely represents analogy or homology, and explain your reasoning: (a) a porcupine’s quills and a cactus’s spines; (b) a cat’s paw and a human’s hand; (c) an owl’s wing and a hornet’s wing.
- WHAT IF?** Suppose that two species, A and B, have similar appearances but very divergent gene sequences, while species B and C have very different appearances but similar gene sequences. Which pair of species is more likely to be closely related: A and B or B and C? Explain.

For suggested answers, see Appendix A.

CONCEPT 26.3

Shared characters are used to construct phylogenetic trees

As we’ve discussed, a key step in reconstructing phylogenies is to distinguish homologous features from analogous ones (since only homology reflects evolutionary history). Next, we’ll describe cladistics, a widely used set of methods for inferring phylogeny.

Cladistics

In the approach to systematics called **cladistics**, common ancestry is the primary criterion used to classify organisms. Using this methodology, biologists attempt to place species into groups called **clades**, each of which includes an ancestral species and all of its descendants. Clades, like the categories of the Linnaean system, are nested within larger clades. In